Reflect Points Across Axes

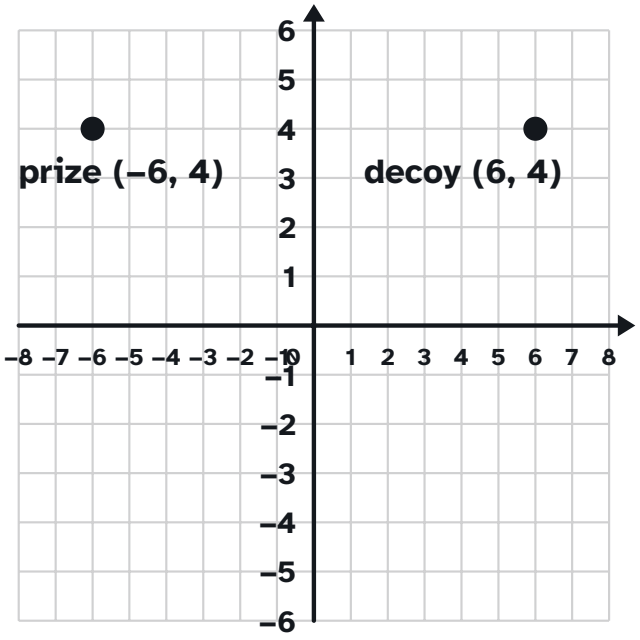
Reflect points across an axis and check the distance from the mirror line.

Name _____ **Date** _____

Tonight's spark

A scavenger hunt map shows a prize at (–6, 4). The host hides a decoy at its mirror twin across the y-axis. How far from the y-axis is the decoy?

Make a guess first: is the decoy closer to the y-axis than the prize, or farther?



Our class model: same distance from the mirror

Try it

Work it out on paper. Write one answer on each line.

1 **Reflect (–4, 2) over the y-axis.**

A. (4, 2) **B.** (–4, –2) **C.** (4, –2) **D.** (–4, 2)

My answer _____

2 **How far is the point (–7, 3) from the y-axis?**

A. 3 units **B.** 7 units **C.** –7 units **D.** 10 units

My answer _____

3 **(5, –2) is reflected over the x-axis. How far is the new point from the x-axis?**

A. 5 units **B.** 3 units **C.** 2 units **D.** 7 units

My answer _____

4 **The point (–9, 4) is reflected over the y-axis. Where is its reflection?**

A. (9, –4) **B.** (–9, –4) **C.** (4, 9) **D.** (9, 4)

My answer _____

5 **A chest is at (6, –1). Its decoy is the reflection over the x-axis. Where is the decoy?**

A. (–6, –1) **B.** (6, 1) **C.** (–6, 1) **D.** (1, 6)

My answer _____

Talk about it

How far is the chest from the mirror axis? How far is the decoy?

Home mission

Use a long table edge or a strip of tape as a mirror line. Place a coin 4 finger-widths from it. Can your student place the coin's mirror twin? Then move the coin closer. What happens to the twin?

If we get stuck, ask

- Which axis is the mirror line here?
- Which coordinate has the same name as that axis?
- Are both points the same distance from the mirror?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/7-9-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it